

SOC Operations & Threat Detection Mastery

CFSS Global Internship 2026 – Final Technical Report

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1. Executive Summary

SOC Operations and Threat Detection Mastery is a Security Operations Center (SOC) monitoring and threat detection project developed during the CFSS Global Internship Program 2026. The project focuses on real-time log monitoring, threat detection, attack simulation, and security analytics using Wazuh SIEM.

The SOC environment successfully identified failed login attempts, brute-force attacks, malware simulation alerts, command execution events, and authentication anomalies.

The project demonstrates practical exposure to SIEM Operations, Log Analysis, Threat Detection, Incident Monitoring, Security Dashboarding, and Blue Team workflows.

2. Project Objective

The objective of this project is to build a functional SOC monitoring environment capable of monitoring system security events, detecting suspicious activities, simulating cyberattacks, and visualizing security analytics using SIEM tools.

3. SOC Methodology

Phase 1 – SOC Lab Setup

- Installed VirtualBox/VMware
- Configured Wazuh SIEM
- Connected monitored endpoints

Phase 2 – Log Collection & Monitoring

- Monitored Windows Event Logs
- Analyzed Event IDs 4624 and 4625
- Categorized logs into security groups

Phase 3 – Threat Simulation & Detection

- Performed brute-force simulations
- Tested malware detection using EICAR
- Investigated command execution logs

Phase 4 – Dashboarding & Reporting

- Built custom SOC dashboard
- Visualized security trends
- Prepared technical documentation

4. Weekly Development Timeline

Week 1 – SOC Lab Deployment

- Installed virtualization environment
- Connected Wazuh agents

Week 2 – Log Analysis & Monitoring

- Monitored authentication logs
- Filtered suspicious events

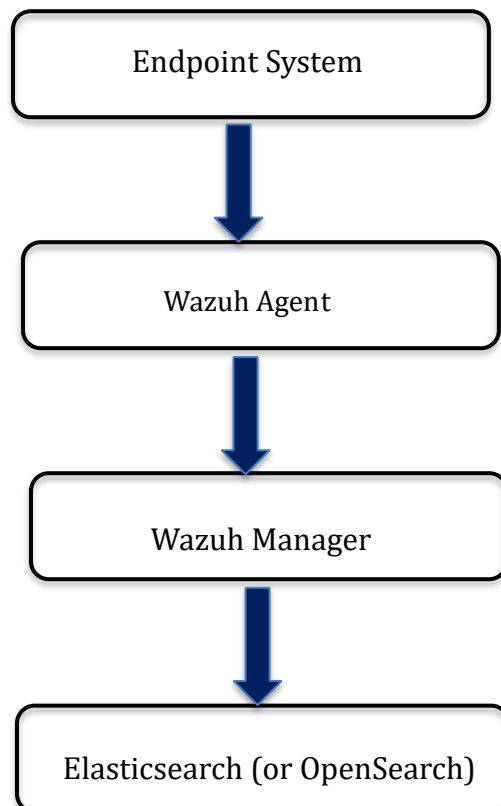
Week 3 – Attack Simulation & Detection

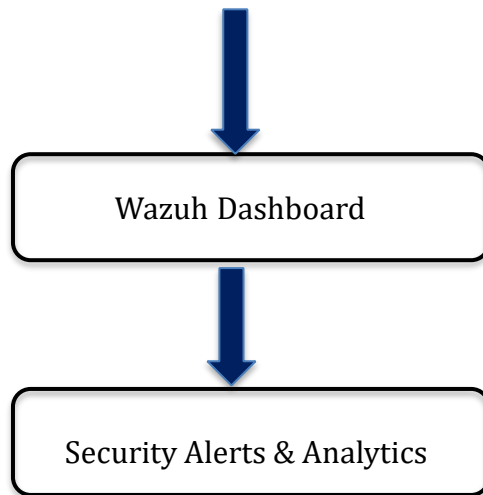
- Simulated brute-force attacks
- Triggered SIEM alerts
- Tested malware detection

Week 4 – Dashboarding & Reporting

- Designed custom dashboard - Compiled final technical report

5. SOC Architecture





6. Technology Stack

SIEM & Monitoring:

- Wazuh SIEM
- Wazuh Dashboard

Operating Systems:

- Windows
- Linux

Virtualization:

- VMware Workstation
- VirtualBox

Log Sources:

- Windows Event Logs
- Syslog

7. Module Explanation

Agent Monitoring Module:

- Endpoint connectivity monitoring
- Log forwarding

Authentication Monitoring Module:

- Successful and failed login monitoring

Threat Detection Module:

- Brute-force detection
- Malware simulation alerts
- Command execution monitoring

Dashboard Analytics Module:

- Daily security alerts
- Alert severity visualization

8. Log Analysis & Threat Detection

Important Event IDs:

- 4624 – Successful Login
- 4625 – Failed Login
- 4688 – Process Creation
- 1102 – Audit Log Cleared

Threat Indicators:

- Multiple failed login attempts
- Unauthorized command execution
- Malware signature detection

9. Attack Simulation Results

Brute Force Detection:

Multiple incorrect password attempts triggered high-severity alerts.

Malware Simulation:

EICAR test file detection successfully generated SIEM alerts.

Command Monitoring:

Commands like 'whoami' and 'net user' were logged successfully.

10. Security Monitoring Dashboard

The dashboard visualized daily alerts, top triggered security rules, agent status, and authentication trends to improve threat visibility and SOC awareness.

11. Technical Findings

- Wazuh agents maintained stable communication
- Failed logins triggered alerts successfully
- Malware simulation detection worked correctly
- Dashboard visualizations updated dynamically
- Security monitoring improved system visibility

12. Security Recommendations

- Enable Multi-Factor Authentication (MFA)
- Integrate email/SMS alerts
- Add threat intelligence feeds
- Configure active response automation
- Implement advanced SIEM correlation rules

13. Challenges & Solutions

Challenge: Agent connection failures

Solution: Reconfigured authentication settings

Challenge: Missing alerts

Solution: Updated Wazuh rule severity

Challenge: Dashboard performance delay

Solution: Optimized indexing configuration

14. Future Enhancements

- AI-based anomaly detection
- Cloud security monitoring
- Endpoint Detection & Response (EDR)
- Threat intelligence integration
- Advanced malware analytics

15. Conclusion

Project on SOC Operations & Threat Detection Mastery successfully demonstrates a real-world SOC monitoring environment capable of detecting and analyzing cybersecurity threats.

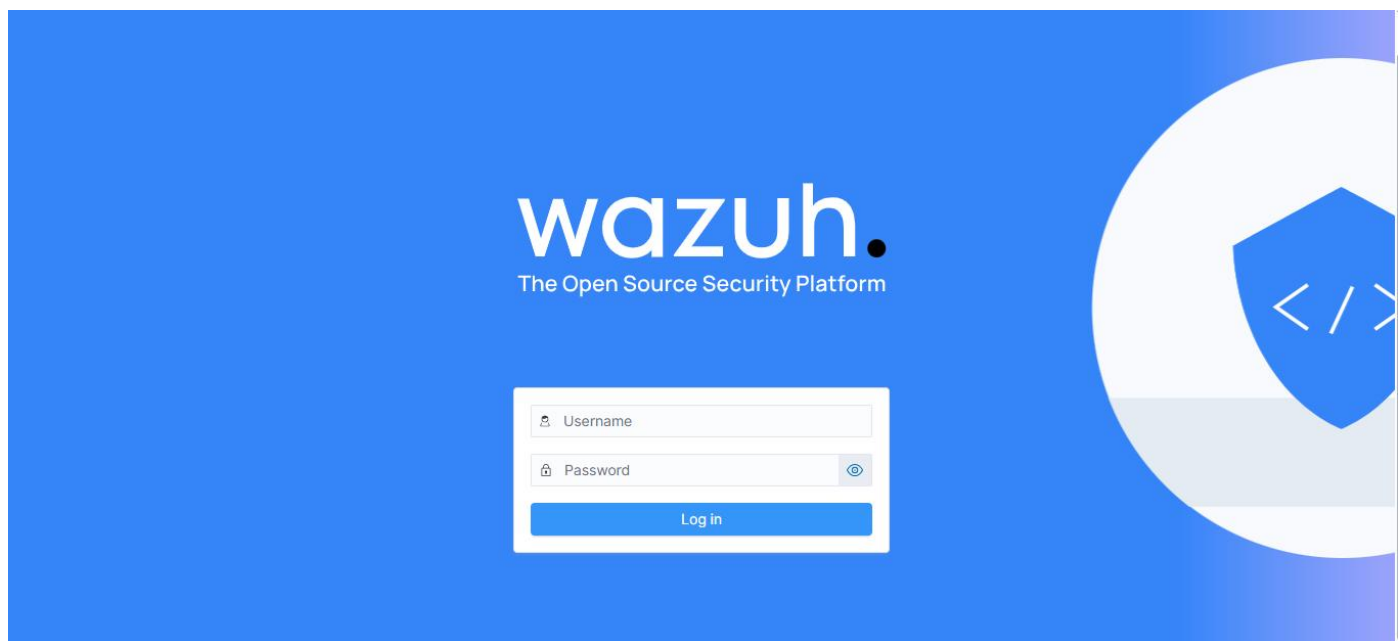
The project provided practical exposure to SIEM operations, threat detection, attack simulation, and security dashboarding, creating a strong foundation for SOC Analyst and Blue Team roles.

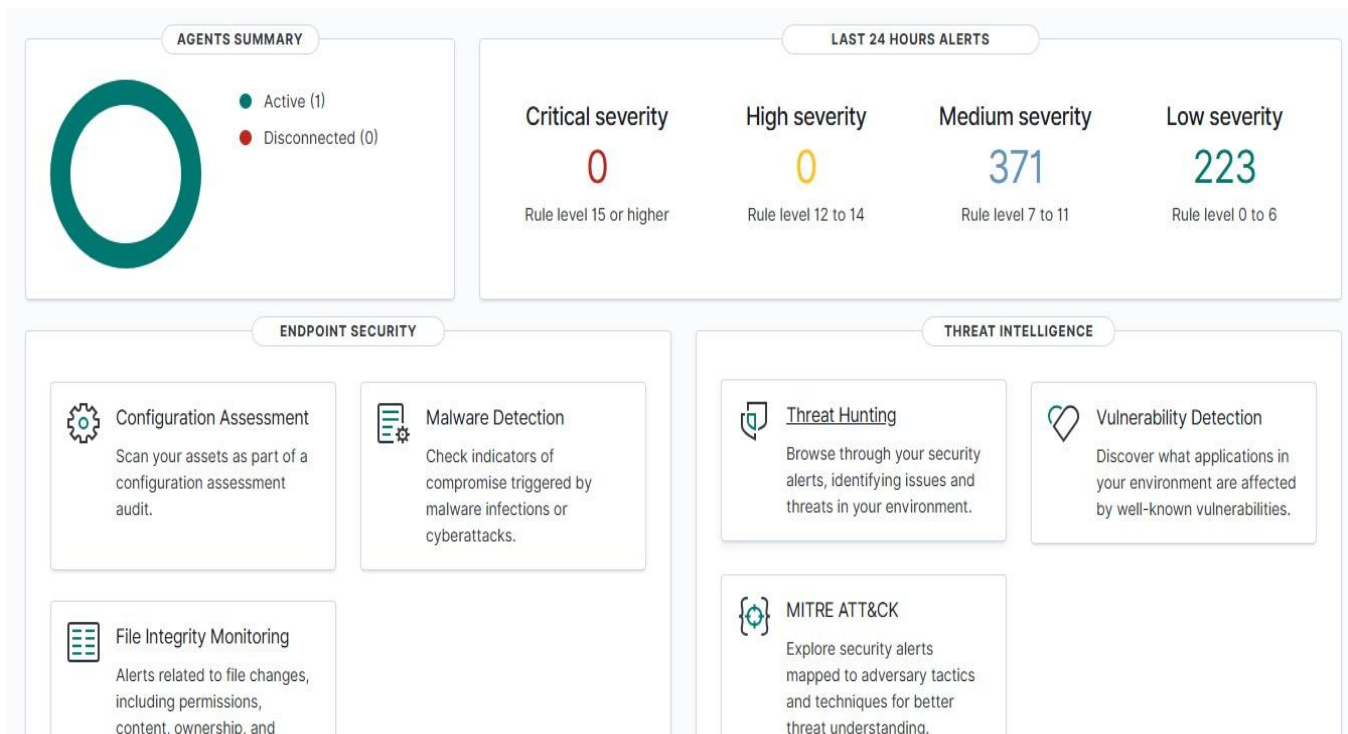
16. Screenshots Appendix

Included screenshots for:

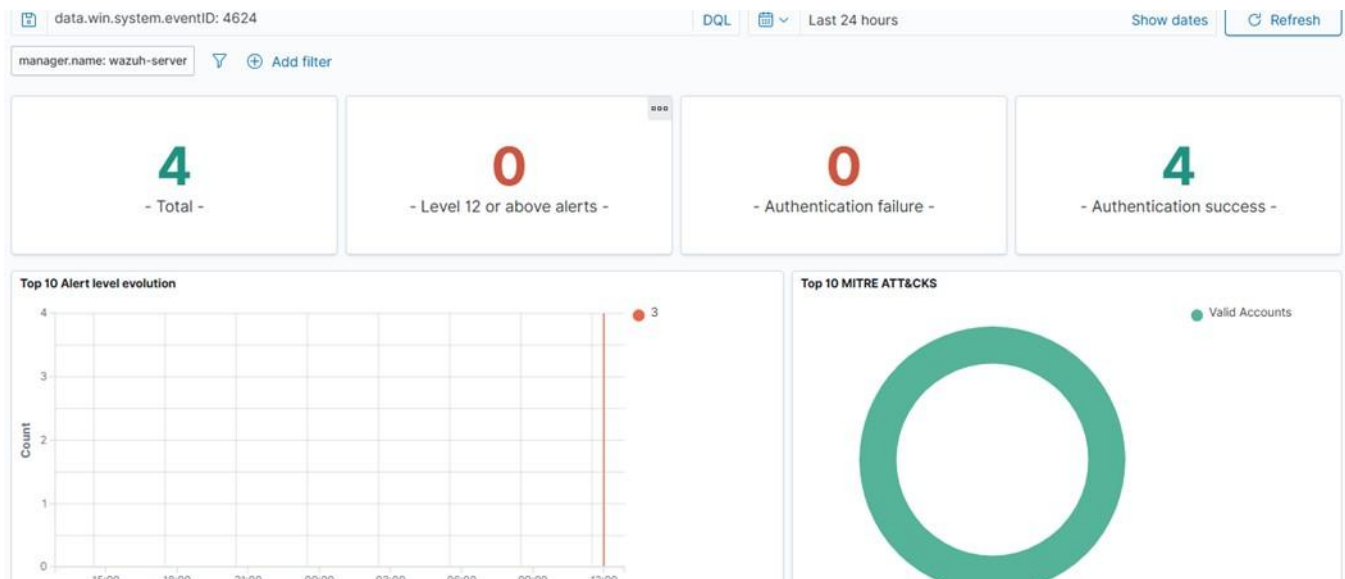
- Wazuh Dashboard and Active Agents
- Account Login and Failed Login Alerts
- Brute Force Alerts Malware Detection Alerts
Command Monitoring Logs
- Wazuh Analytics Dashboard. (Total Alerts per day, Top 5 Security Rules Triggered, Agent Status)

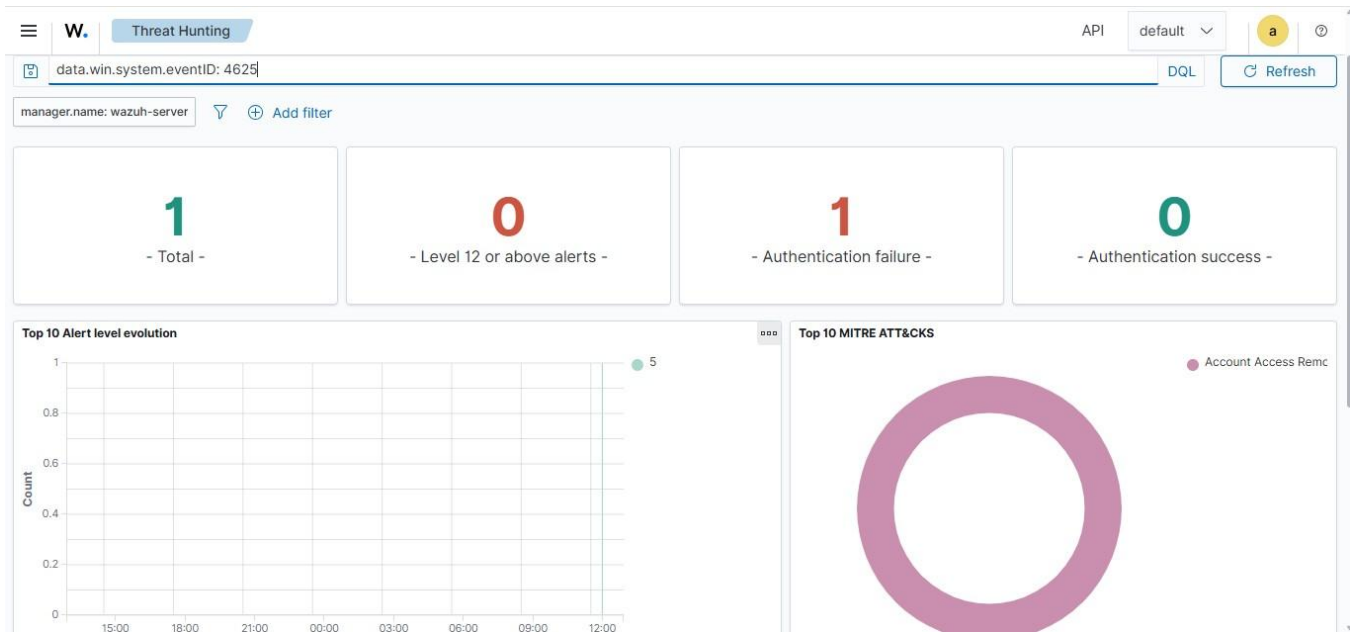
Wazuh Dashboard and Active Agents:





Account Login and Failed Login Alerts:





Brute Force Alerts:

	timestamp	rule.description	rule....	rule.id
	May 1, 2026 @ 15:53:13.309	Logon Failure - Unknown user or bad password	5	60122
	May 1, 2026 @ 15:53:13.327	Logon Failure - Unknown user or bad password	5	60122
	May 1, 2026 @ 15:53:13.311	Logon Failure - Unknown user or bad password	5	60122
	May 1, 2026 @ 15:53:13.312	Logon Failure - Unknown user or bad password	5	60122
	May 1, 2026 @ 15:53:13.330	Logon Failure - Unknown user or bad password	5	60122



info@wazuh.com
<https://wazuh.com>

Alerts summary

Rule ID	Description	Level	Count
60122	Logon Failure - Unknown user or bad password	5	7



info@wazuh.com
<https://wazuh.com>

7

- Total -

0

- Level 12 or above alerts -

7

- Authentication failure -

0

- Authentication success -

Malware Detection Alerts:

Protection history

View the latest protection actions and recommendations from Windows Security.

All recent items

Filters 



Threat removed

01/05/2026 22:31

Severe 

Detected: Virus:DOS/EICAR_Test_File

Status: Removed

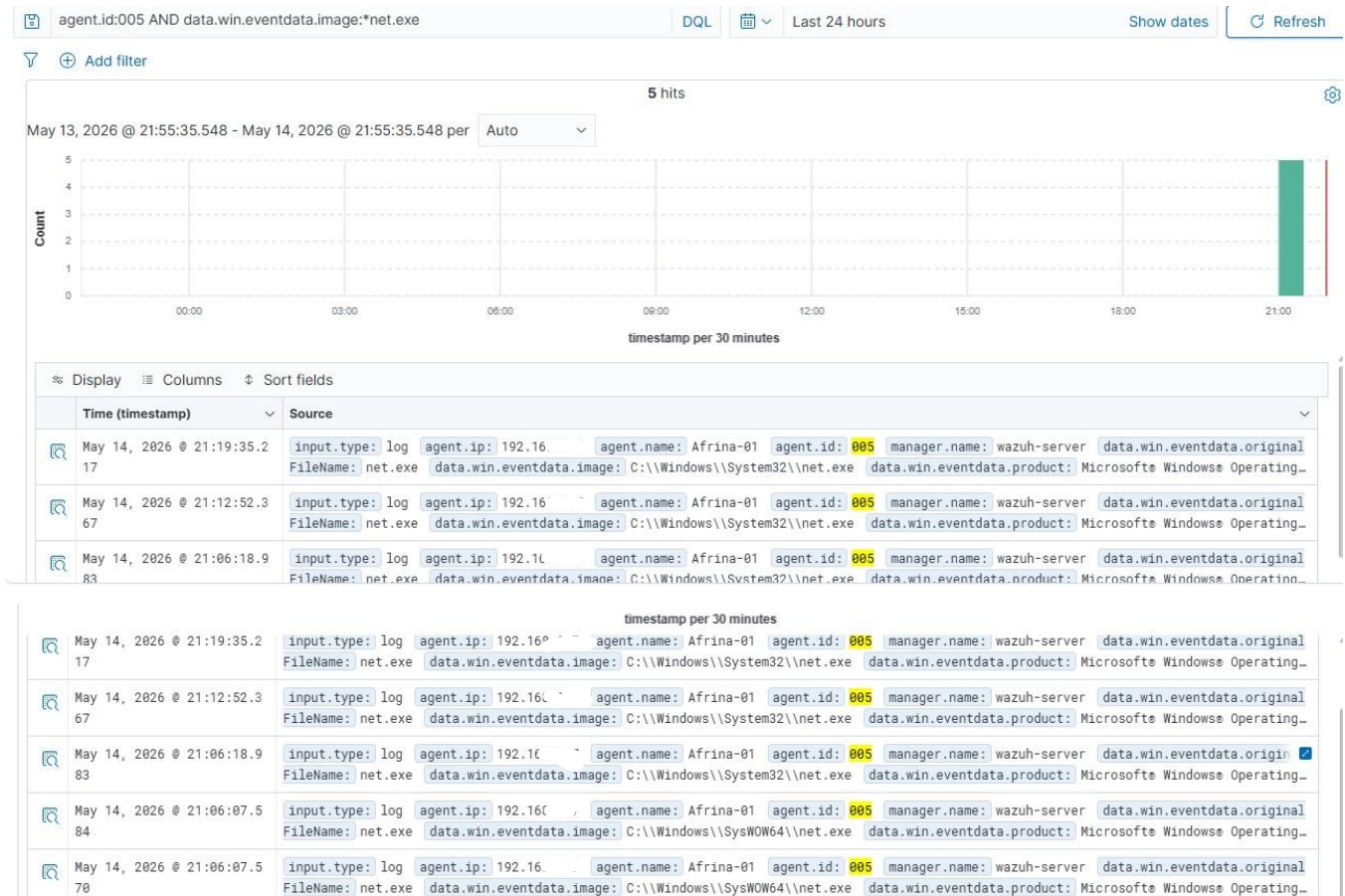
A threat or app was removed from this device.

Date: 01/05/2026 22:32

Details: This program is dangerous and replicates by infecting other files.

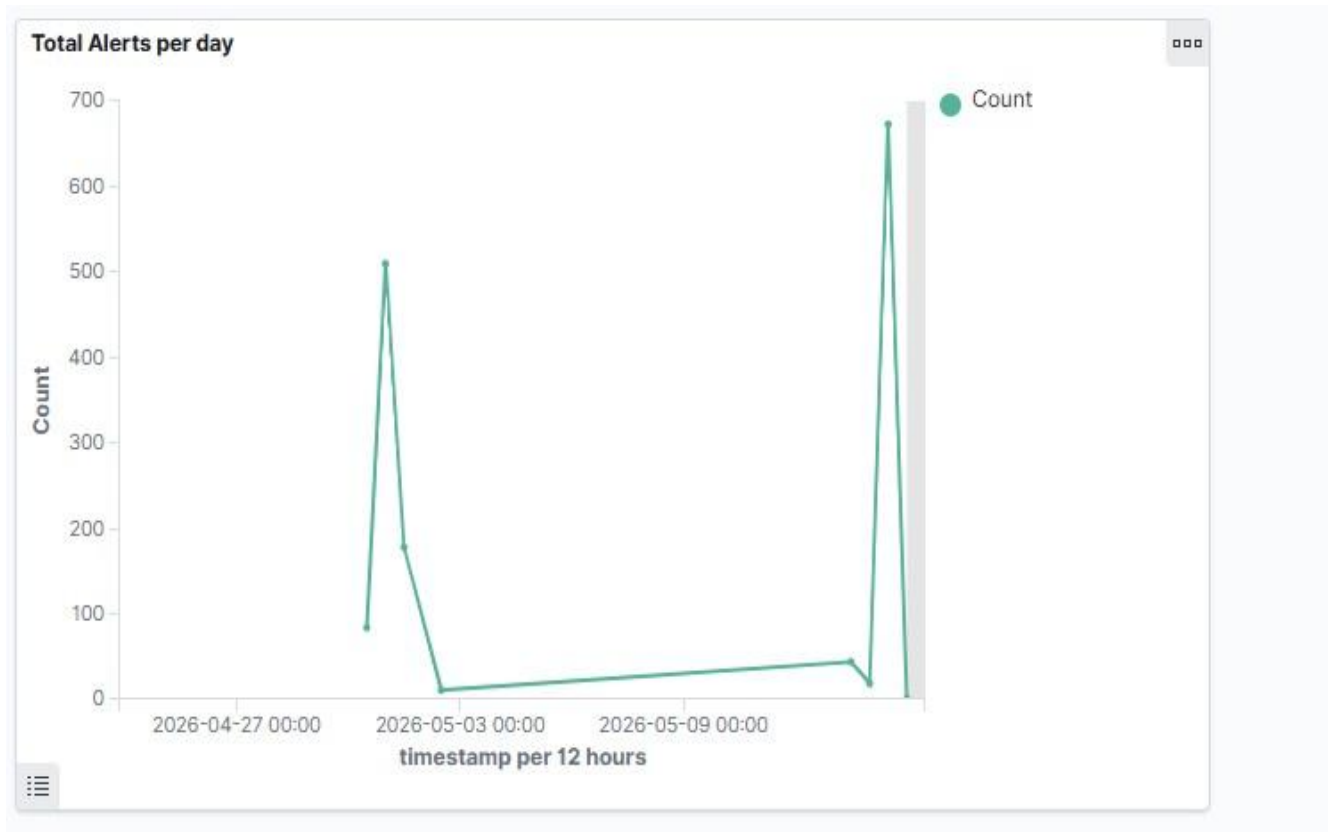


Command Monitoring Logs:

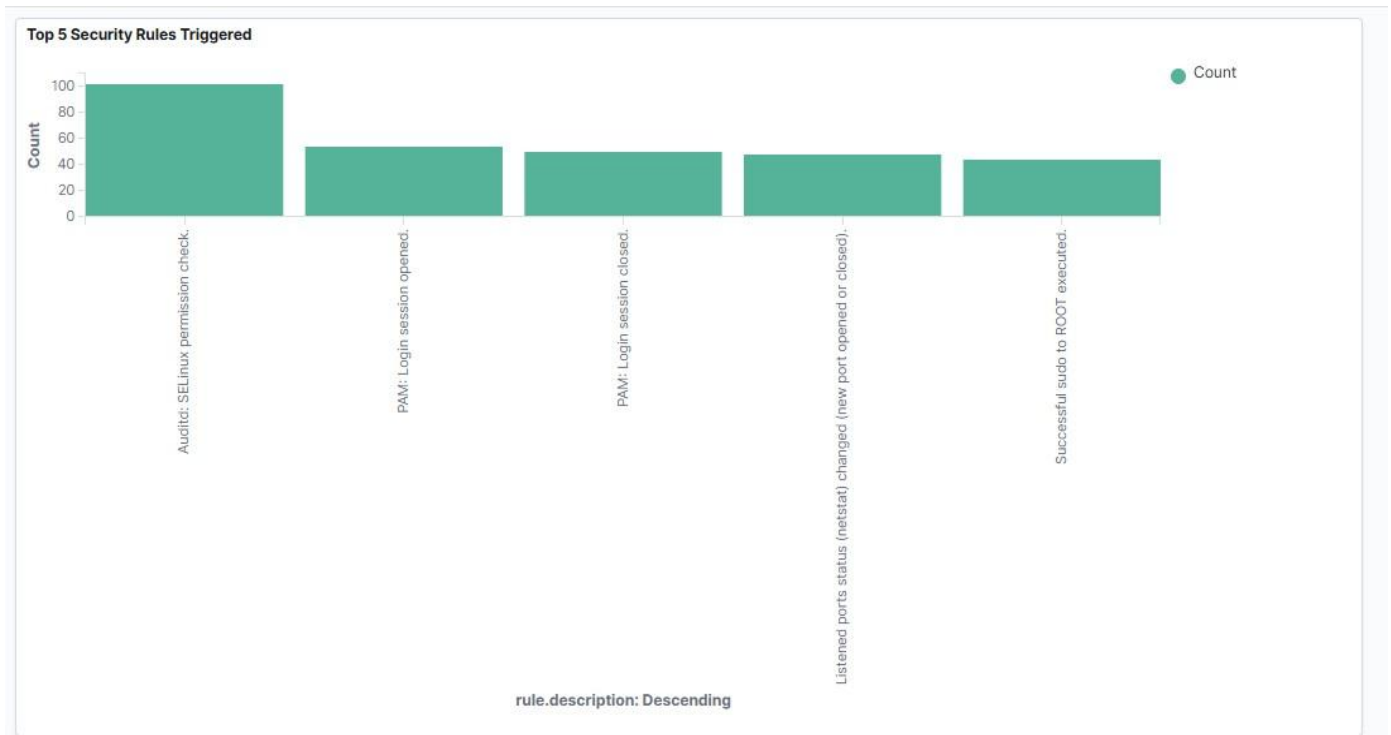


Wazuh Analytics Dashboard

Total Alerts per day:



Top 5 Security Rules Triggered:



Agent Status:

